

Mathematical Foundations of Political Methodology Lecture 1

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Today's outline

- Course context
- Set Theory
- Vectors and Ordered Lists
- Functions

Political Methodology at Chicago

- This Course (PLSC 43401)
 - Calculus/Linear Algebra, Probability/Statistics
 - Followup: PPHA 44200, PLSC 30700, Game Theory I
- Computational Tools for Social Science (R. Terman) (PLSC 31101)
 - Computational techniques, Replicable coding practices.
 - Uses: Everyday programming, writing/publishing papers.
- Social Choice Theory (M. Nalepa) (PLSC 408001)
 - Normative and positive models of decision-making.
 - Uses: understanding societies, micro-foundations for game theory,

Comments on Math and Politics

- This is not what the typical political scientist signed up for.
- Do not skip steps. If you lack high school math, get high school math (not this course).
- Textbooks are your friend.
 - Unlike other classes, the textbook here is necessary.
 - Ideally, 20 pages $\sim$ 2 hours, but even a single page can be an hour.
 - Sometimes you need to make up new problems for yourself, test yourself as you go.
 - If an idea is too complex, find a simpler presentation.
 - We will be having weekly, in class quizzes on the readings (the first is available on Canvas).

A few notes on difficulty

- This course is the broadest offered in the department: we cover 300+ years of mathematical insights. Don't expect your understanding to be immediate, but demand clarification in class.
- “The Owl of Minerva begins its flight at the break of dawn.” -Hegel
- Mathematical training has advantages, hundreds of thousands have gone before you.

Resources

- Math

- Siegel and Moore *A Mathematics Course for Political and Social Research* [LINK](#)
- Khan Academy, edX, MIT OpenCourseWare

- R

- Rochelle Terman's course.
- [datacamp.com](#)
- Wickham *R for Data Science* [LINK](#)
- James, Witten, Hastie and Tibshirani *An Introduction to Statistical Learning with Applications in R*

- Probability/Statistics/Econometrics Books

- Ash, *The Probability Tutoring Book*
- Angrist Pischke *Mostly Harmless Econometrics*

Course Practices

- Using the course materials:
 - Read the textbook and the lecture notes before and after class.
 - The textbooks are great, go back to them often.
- Team/Group learning:
 - The best learning is explaining things to each other.
 - Your goal: to be able to explain what you learn to others.
 - Assignments can be worked on in groups, but all answers must be your own.
- Practice, practice, practice
 - We will be offering practice problems every week related to the course.
 - Our TA, Arthur, will further provide worked examples.
 - My office hours are available as well, from 11:00-12:00 on Tuesdays and Thursdays.

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- Functions

Why Set Theory

- “The task of philosophy is to break the domination of words over the human mind.” - Frege
- The project of Set Theory was to root mathematics in pure logic, it is now more of a background language.
- Set theory allows us to map real world stuff into the language of mathematics.

Mathematical Set

Definition

A set is a collection of objects viewed as a single entity.

- If the objects in the set can be counted then it's common for the set to be written by enclosing the objects in curly braces $\{\dots\}$
 - The curly braces are extremely important: $\{1, 4, 5\} \neq (1, 4, 5)$
- Examples of sets: $\{\text{coffee, Bobby}\}$, or $\{2, 5, 3, 8, 10\}$, or $\{\text{red, green, blue}\}$
- The elements of a set must all be unique: $\{1, 1, 3, 4, 4\}$ is not a set
- The ordering of elements in a set does not matter:
 $\{\text{red, blue, green}\} = \{\text{green, blue, red}\}$

An aside on Mathematical systems

- Mathematics is particular about what can and cannot be done, and what is and is not a thing.
- The algebra of sets defines the properties, laws and relations among sets.
- Common rules like the associative property of multiplication used in middle school do not hold across contexts.

Contents of sets: Logical 'in'

Definition

Is an element of: $\in$.

Examples:

- $\text{green} \in \{\text{red}, \text{green}, \text{blue}\}$
- $\text{magenta} \notin \{\text{red}, \text{green}, \text{blue}\}$
- In R we can test this with `%in%`.
- $\in$ should only be used to relate elements of sets to sets.

Test Your Understanding

- $\text{red} \in \{\text{red}, \text{green}, \text{blue}\}$
- $\{\text{green}, \text{blue}\} \in \{\text{red}, \{\text{green}, \text{blue}\}\}$
- $\{\text{red}\} \in \{\text{red}, \text{green}, \text{blue}\}$
- $\emptyset \in \{\text{red}, \text{green}, \text{blue}\}$

Test Your Understanding

- $\text{red} \in \{\text{red}, \text{green}, \text{blue}\}$ TRUE
- $\{\text{green}, \text{blue}\} \in \{\text{red}, \{\text{green}, \text{blue}\}\}$ TRUE
- $\{\text{red}\} \in \{\text{red}, \text{green}, \text{blue}\}$ FALSE
- $\emptyset \in \{\text{red}, \text{green}, \text{blue}\}$ TRUE

Relations between sets: Subset

Definition

Is a subset of: $\subset$.

Examples:

- $\{\text{green}, \text{red}\} \subset \{\text{red}, \text{green}, \text{blue}\}$
- Note that $\text{green} \not\subset \{\text{red}, \text{green}, \text{blue}\}$ because “green” IS NOT a set!
- However $\{\text{green}\} \subset \{\text{red}, \text{green}, \text{blue}\}$ because $\{\text{green}\}$ IS a set

Remark:

- Sometimes a distinction is made between $\subset$ (strict subset) and $\subseteq$ (weak subset); the former excludes being the whole set.

Relations between sets: Union

Definition

Union is $\cup$, and represents the set of elements in either set.

Examples:

- $\{\text{Arthur, Bobby}\} \cup \{2, 5, 3, 8, 10\}$
 $= \{\text{Arthur, Bobby, 2, 5, 3, 8, 10}\}$
- $\{\text{Arthur, Bobby}\} \cup \{2, 5, 3, 8, 10\} \cup \{\text{red, green, blue}\}$
 $= \{\text{Arthur, Bobby, 2, 5, 3, 8, 10, red, green, blue}\}$
- $\{\text{Arthur, Bobby}\} \cup \{\text{Arthur, Bobby}\} = \{\text{Arthur, Bobby}\}$

Relations between sets: Intersection

Definition

Intersection is $\cap$, and represents the set of elements in both sets.

Examples:

- $\{\text{Arthur, Bobby}\} \cap \{\text{Bobby, Pemberton}\} = \{\text{Bobby}\}$

- $\{\text{Arthur, Bobby}\} \cap \{2, 5, 3, 8, 10\} = \emptyset$

$\emptyset$ is the “empty set” which is the set that contains nothing, or $\{\}$ The output of union and intersection operations *is always a set!*

- $\{\text{Arthur, Bobby}\} \cap \{\text{Bobby, Pemberton}\} \neq \text{Bobby}$

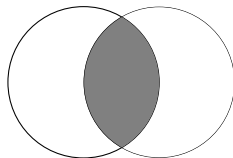
If two sets have an empty intersection, they are *disjoint*

- $\{\text{Arthur, Bobby}\}$ and $\{2, 5, 3, 8, 10\}$ are disjoint

Useful Heuristics

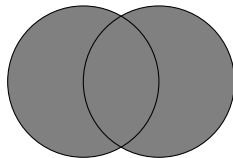
- $\cap$, or the set **intersection**, is related to the logical **and** ($\&$).

Intersection



- $\cup$, or the set **union**, is related to the logical **or** ($\vee$).

Union



Set Rules

- Commutative rules:

- $A \cup B = B \cup A$;
- $A \cap B = B \cap A$;

- Associative rules:

- $(A \cup B) \cup C = A \cup (B \cup C)$;
- $(A \cap B) \cap C = A \cap (B \cap C)$;

- Distributive rules:

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;

Concise Notation for Set

Oftentimes we don't want to (or can't) simply list all the items in a set, so we define sets in other ways (note we *always use the curly braces*)

- The set of all the boys' names that start with a vowel:
 $\{x : x \text{ is a boy's name starting with a vowel}\}$
- The set of all real numbers greater than -17.5 and less than 4: $\{x \in \mathbb{R} : -17.5 < x < 4\}$

Using this notation, we can define the following:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Note, $:$ means 'such that'

Family of Sets and Cardinality of Set

- A *family of sets* is a set whose elements are themselves sets
 - $\{\{1, 2, 4\}, \{2, 6\}, \{\text{green}\}\}$
- A well-known family of sets is the *power set* of any set X : it is the set of all subsets of X , and is written 2^X
 - How many ways can we create sets of $\{\text{cat}, \text{dog}\}$?
 - $2^{\{\text{cat}, \text{dog}\}} = \{\emptyset, \{\text{cat}\}, \{\text{dog}\}, \{\text{cat}, \text{dog}\}\}$
- $|X|$ is the cardinality (size) of set X ; the number of elements in X
 - If $A = \{\text{dog}, \text{cat}\}$, $|A| = 2$
 - If $|X| = n$ then $|2^X| = 2^n$

Complementary Set Properties

- If $A \subset B$ then the set of things that are in B but not A is the *complement of A in B*

$$B \setminus A = \text{The complement of } A \text{ in } B = \{x \in B : x \notin A\} = B \cap A^c$$

- De Morgan's laws:
 - $(A \cup B)^c = A^c \cap B^c$.
 - $(A \cap B)^c = A^c \cup B^c$.

Important Sets of Numbers

- A **finite** set contains a finite number of elements.
- An **infinite** set contains an infinite number of elements.
- The set of real numbers, $\mathbb{R}$: All numbers that are positive, negative, or 0
- The set of natural numbers, $\mathbb{N}$: The set of all strictly positive integers
 $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- The set of integers, $\mathbb{Z}$: the natural numbers, their negatives, and 0
 $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

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Why Vectors and Ordered Lists

- Sets ignore order, but we often have a collection of data where each value corresponds to a feature.
- For example, my pet has (at least) three attributes: a species (dog) a gender (female) and a name (Maeby).
- The set {dog, female, Maeby} would lose this ordering.
- The vector and ordered list saves these placeholder data.
- In our work, we will see most data is in the form of either a vector (if it is pure numbers) or an ordered list (if it is a mix of kinds of data).

Ordered List

- Unlike sets, *ordered lists* are denoted with smooth brackets (c, a, b, d, e)
 - Elements can repeat
 - These are sometimes called sequences.
- The 2-dimensional plane is represented by $\mathbb{R}^2$, or $\mathbb{R} \times \mathbb{R}$

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$$

- The 3-dimensional space is represented by $\mathbb{R}^3$:

$$\mathbb{R}^3 = \{(x, y, z) : x \in \mathbb{R} \text{ and } y \in \mathbb{R} \text{ and } z \in \mathbb{R}\}$$

Cartesian Product

- Ordered pairs can be generalized to any two objects, the possibilities are given by the Cartesian Product.
- Given two sets X and Y , the Cartesian product of X and Y is $\{(x, y) : x \in X \text{ and } y \in Y\}$
 - $\{\text{cat}, \text{dog}\} \times \{a, b\} = \{(\text{cat}, a), (\text{cat}, b), (\text{dog}, a), (\text{dog}, b)\}$
 - $X_1 \times X_2 \times \dots \times X_n = \{(x_1, x_2, \dots, x_n) : x_i \in X_i \text{ for } i = 1, 2, \dots, n\}$
- $\mathbb{R}^n$ is the n -ary Cartesian product of $\mathbb{R}$

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}$$

- $\mathbb{R}^2$ is the typical 2d plane
- $\mathbb{R}^3$ is the 3d space

Vector

Definition

An ordered list of numbers is a vector.

If the vector has n elements, then it is an element of n -dimensional Euclidean space, or $\mathbb{R}^n$

Examples:

- $(1, 5, 7) \in \mathbb{R}^3$
- $(3, 2, 7, 12, 65, 78787, 3) \in \mathbb{R}^7$

Remark:

- If $n \leq 3$, you can plot out the vector (as either on a line, the plane, or 3-D space).
- In R, we write a vector with `c()`:
`c( 25, 30, 40)`

Analogy from data to coordinates

- It is useful to be able to think about vectors either as an ordered list of numbers or as a point in space.
- The analogy to space offers insights, but is limited to vectors of length less than 3.
- The data encoded in a length 2 vector (a, b) can be used to describe an arrow from the origin to the point $x = a, y = b$

Outline

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Why Functions?

- Functions describe regularity: the fundamental target of science.
- Theoretical work relies on functions to describe relationships.
- Random variables are functions (more on that later).
- Here we just introduce some basic terminology.

Mathematical Functions

Definition

A function is a mapping from a set X into a set Y

- This means that for each element of X , f assigns an element of Y , or $f : X \rightarrow Y$
- The *graph* of a function from $\mathbb{R}$ to $\mathbb{R}$ is the set

$$\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$$

Aside on variables

Definition

The letters X, Y, A, B, x, y are variables.

- Common to use script upper and lower case letters.
- We will also use the Greek letters.
alpha α ; beta β ; gamma γ Γ ; delta δ Δ ; epsilon ϵ ; zeta ζ ; eta η ; theta θ Θ ; iota ι ;
lambda λ Λ ; mu μ ; nu ν ; omicron $\omicron$; pi π Π ;
rho ρ ; sigma σ Σ ; tau τ ; upsilon υ Υ ; phi ϕ Φ ; chi χ ;
psi ψ Ψ ; omega ω Ω
- As a last resort, we go to the Hebrew (אבגדה)

Classes of Functions

Definition

A function $f : X \rightarrow Y$ is one-to-one (or injective) if every element of X maps into a unique element of Y ($f(a) = f(b) \Rightarrow a = b$)

Definition

A function $f : X \rightarrow Y$ is onto (or surjective) if every element of Y is mapped into by some element of X ($\forall y \in Y, \exists x \in X : f(x) = y$)

Definition

A function that is one-to-one and onto is bijective

Aside on logic terms

- For all $\forall$, exists $\exists$, if and only if $\iff$
 - $\forall y \in Y, \exists x \in X : f(x) = y$: “For all y in the set Y , there exists an x in the set X such that the function f evaluated at x is equal to y .”
 - $\nexists a \in Y \iff \forall y \in Y, y \neq a$: There does not exist an element a in the set Y if and only if for all elements y in the set Y , y is not equal to a .

Test Your Understanding

- $f : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^2$ is injective and surjective
- $g : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^3$ is surjective
- $g : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^3 - 3x$ is injective

Test Your Understanding

- $f : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^2$ is injective and surjective FALSE, neither is true.
- $g : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^3$ is surjective TRUE.
- $g : \mathbf{R} \rightarrow \mathbf{R} \ f(x) = x^3 - 3x$ is injective FALSE, it is surjective.

Composite Functions

Functions can be combined. When they are, they are called composite functions:

Definition

If $f : B \rightarrow C$ and $g : A \rightarrow B$ then

$f \circ g = f(g(x))$ is a mapping from A to C : $x \in A$, $g(x) \in B$, and $f(g(x)) \in C$

Composite Function Example

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with $f(x, y) = x + 2y$
- $g : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, with $g(x, y, z) = (x^2 + 2y * z, 7)$

■

$$h(x, y, z) = f \circ g = f(g(x, y, z), y) : \mathbb{R}^3 \rightarrow \mathbb{R}, \text{ with}$$

$$h(x, y, z) = x^2 + 2y * z + 14$$

In-class problems

3.1.2 Let A, B, C be sets such that $A \subset B$ and $A \cap C = \emptyset$. Simplify the following, if possible:

- $A \cap (B \cup C)$
- $(A \cap B) \cup C$
- $A \cup (B \cap C)$
- $(A \cup B) \cap C$

In-class problems

- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Interpret the mapping $f(x, y) = (x, -y)$ geometrically
 - Which points remain unchanged under this mapping?
 - Now answer both questions for $f(x, y) = (x, -4)$

Important things from Pemberton and Rau

- How to sketch functions.
- Linear programming.
- Real, rational, irrational numbers.
- Functions